13.5.6 Wrap-Up: Ticket Out the Door

1. A number line is shown. Each grid space is 1 unit.

a. Determine the point on the number line that satisfies the expression shown. Label the point M.

$$\frac{2}{5}(\text{point } W)+\frac{3}{5}(\text{point } T)$$

b. What is the ratio $WM:MT$?

Unit 13, Lesson 6: Classifying Triangles

Benchmark

MA.912.GR.3.2 Given a mathematical context, use coordinate geometry to classify or justify definitions, properties, and theorems involving circles, triangles, or quadrilaterals.

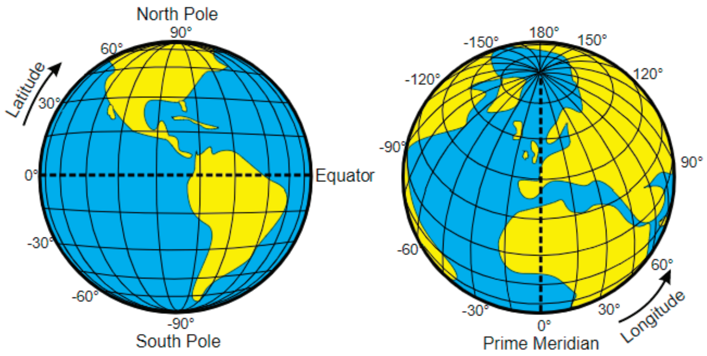
Learning Targets

- ☐ I can use coordinate geometry to classify triangles in a mathematical or real-world context.
- ☐ I can draw a triangle on a coordinate plane using given definitions, properties, or theorems.

13.6.1 Warm-Up: Latitude and Longitude

In ancient times, people used landmarks to identify locations, but travelers were not familiar with local landmarks, making traveling difficult. Hipparchus, a Greek astronomer, was the first to specify location using latitude and longitude as coordinates. Hipparchus applied his knowledge of spherical angles to the problem of denoting locations on the Earth's surface.

Latitude is the measurement of the globe north to south, and longitude is the measurement of the globe east to west.



The table shows the latitude and longitude of different Florida cities.

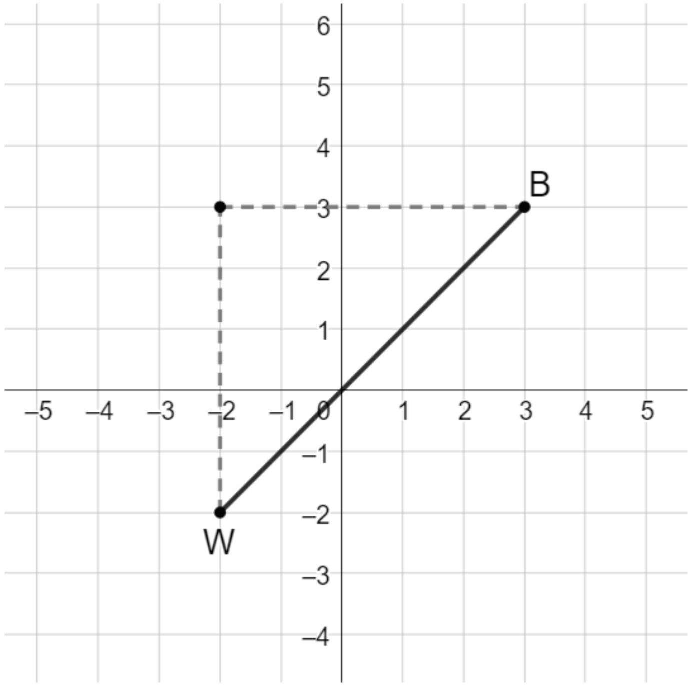
City	Coordinates	City	Coordinates
Panama City	$30.2N, 85.7W$	Miami	$25.8N, 80.2W$
Tallahassee	$30.46N, 84.28W$	Tampa	$28.0N, 82.5W$
Lake Worth	$26.6N, 80.1W$	Orlando	$28.5N, 81.4W$
Jacksonville	$30.3N, 81.7W$	Gainesville	$29.7N, 82.3W$

1. What do you notice? What do you wonder?

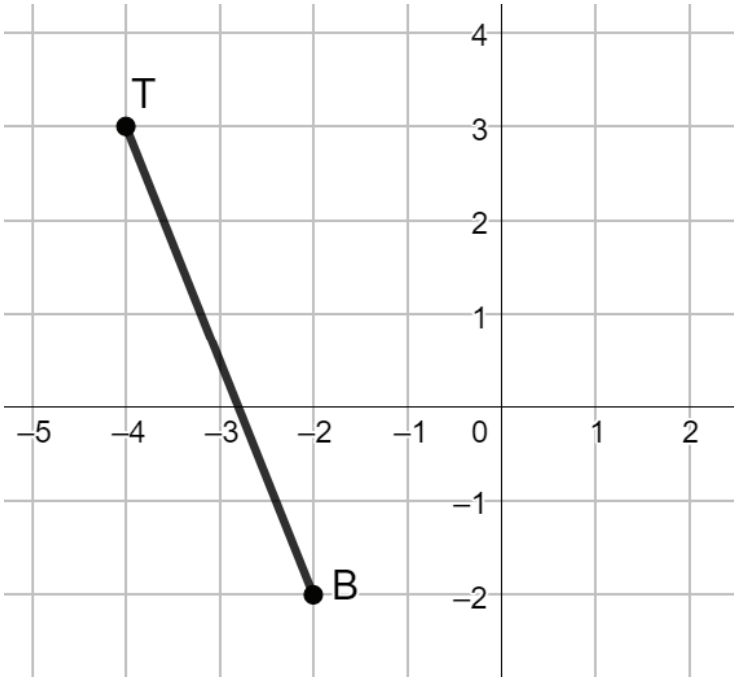
13.6.2 Refresher: Finding Distance

Work with your partner to complete the following.

1. Find the length of $\overline{WB}$ using the Pythagorean Theorem. If necessary, round to the nearest thousandth.



2. Use the Pythagorean Theorem to find the length of $\overline{TB}$. If necessary, round to the nearest thousandth.

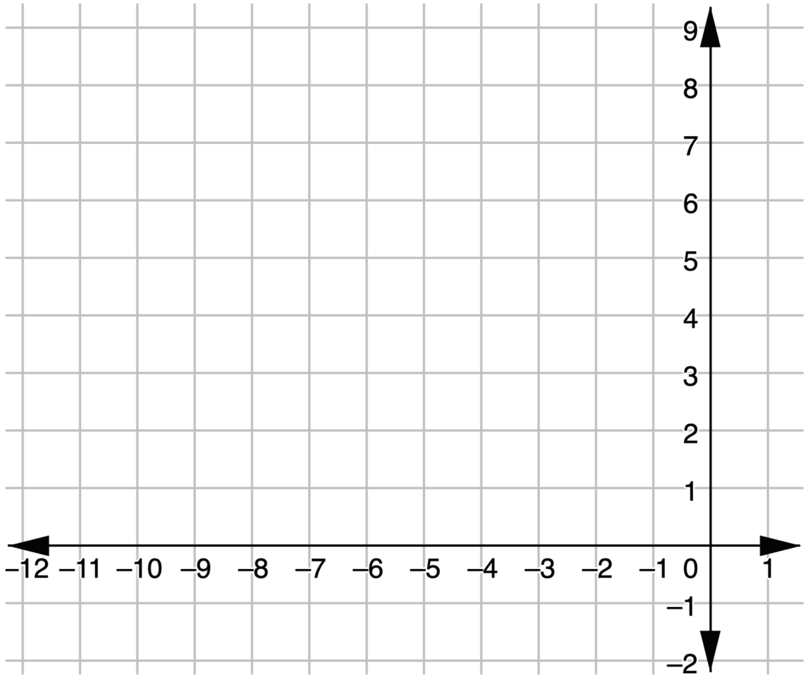


3. Explain how the Pythagorean Theorem can be used to find the distance between two points.

13.6.3 Exploration: Classifying Triangles

1. ΔMKC has vertices at $M(-8,8)$, $K(-4,2)$, and $C(-9,3)$.

   a. Graph ΔMKC on the coordinate plane.



   b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{CM}$	
$\overline{MK}$	
$\overline{KC}$	

c. What do you notice about the side lengths of ΔMKC?

d. Complete the statement.

ΔMKC is classified as a(n)
- ○ equilateral
- ○ isosceles
- ○ scalene

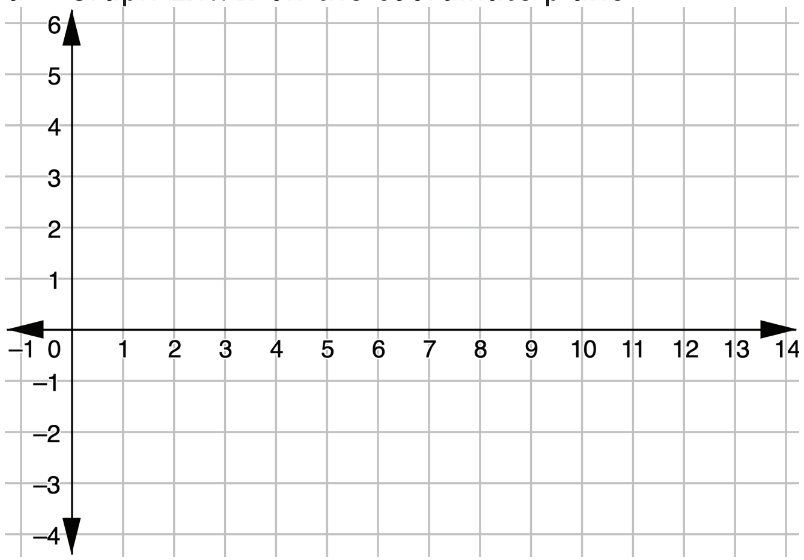
triangle

because ΔMKC has
- ○ no
- ○ two
- ○ three

equal side lengths.

2. ΔNWR has vertices at $N(6,5)$, $W(3,-1)$, and $R(10,-1)$.

a. Graph ΔNWR on the coordinate plane.



b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{NR}$	
$\overline{RW}$	
$\overline{WN}$	

c. What do you notice about the side lengths of ΔNWR?

d. Complete the statement.

ΔNWR is classified as a(n)
- ○ equilateral
- ○ isosceles
- ○ scalene

triangle

because ΔNWR has
- ○ no
- ○ two
- ○ three

equal side lengths.

3. ΔMSG has vertices at $M(-3,4)$, $S(2.2,1)$, and $G(-3,-2)$.

a. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	Length of Line Segment
$\overline{MS}$	
$\overline{SG}$	
$\overline{GM}$	

b. Complete the statement.

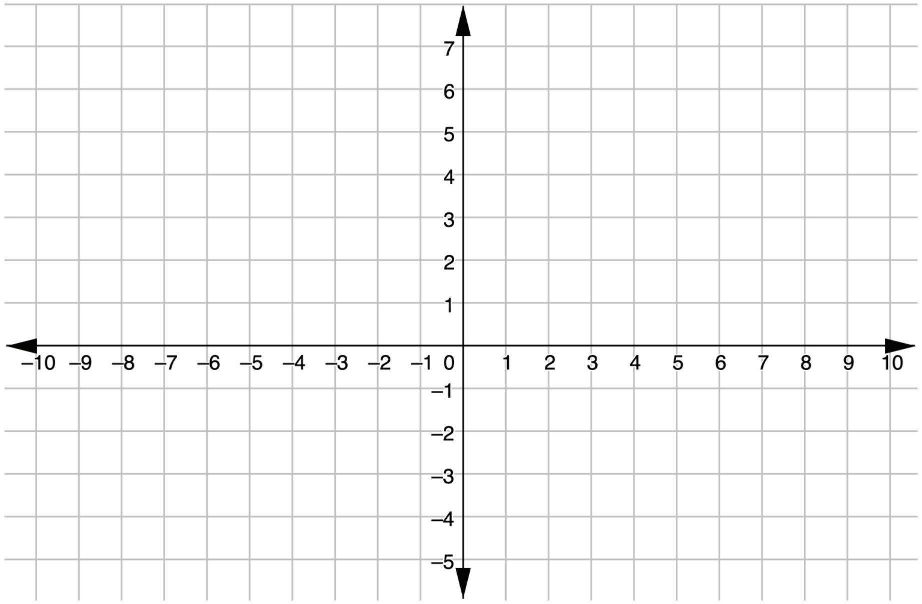
ΔMSG is classified as a(n)
- ○ equilateral
- ○ isosceles
- ○ scalene

triangle

because ΔMSG has
- ○ no
- ○ two
- ○ three

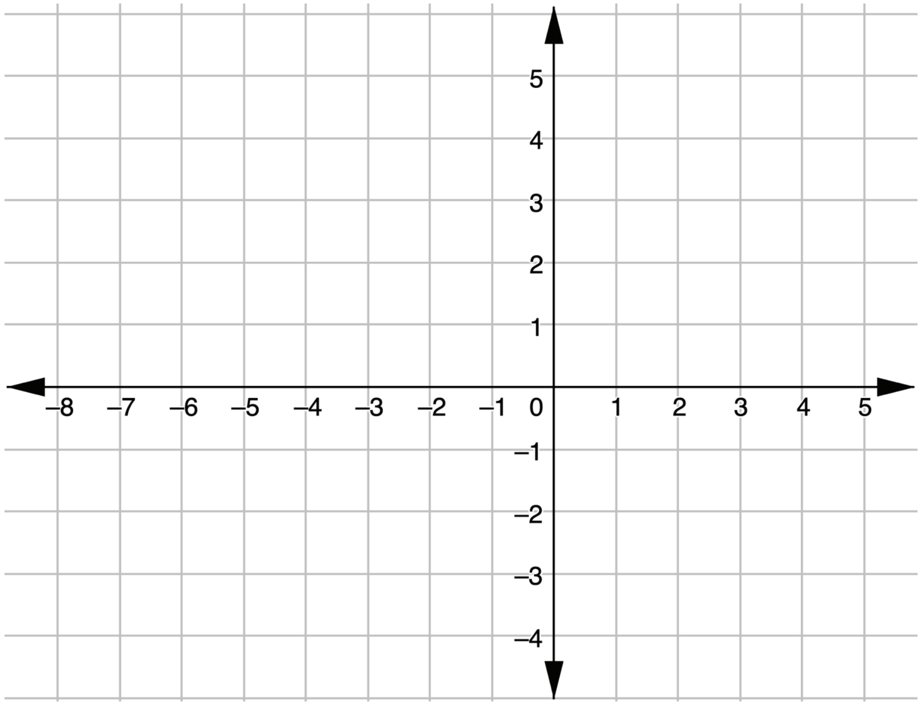
equal side lengths.

4. Draw a scalene triangle with one vertex at $(-3,6)$ and with a base that is twice the length of its height.



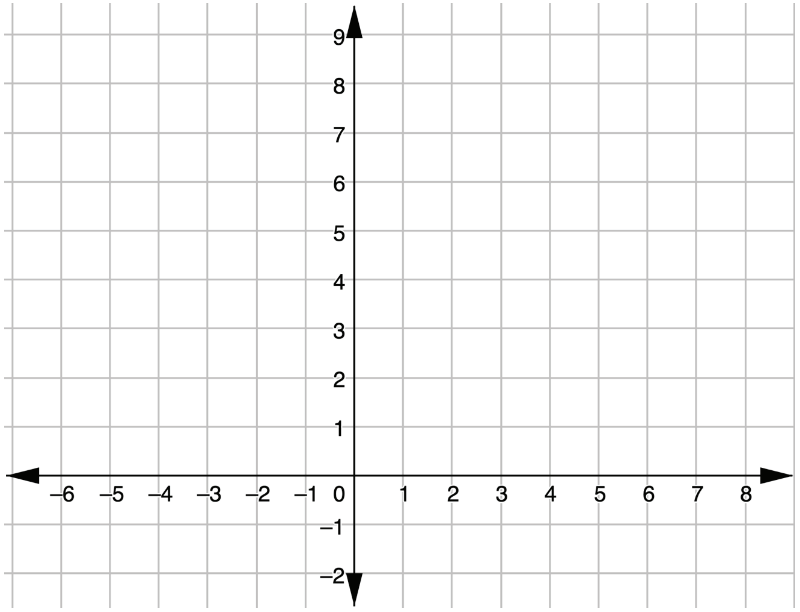
13.6.4 Your Turn!

1. ΔPTH has vertices at $P(-2,3)$, $T(3,1)$, and $H(0,-2)$. Determine if ΔPTH is an equilateral, isosceles, or scalene triangle. Justify your answer with math.



2. Given: ΔSCW with $W(1,6)$, $S(7,6)$, and $C(4,0)$

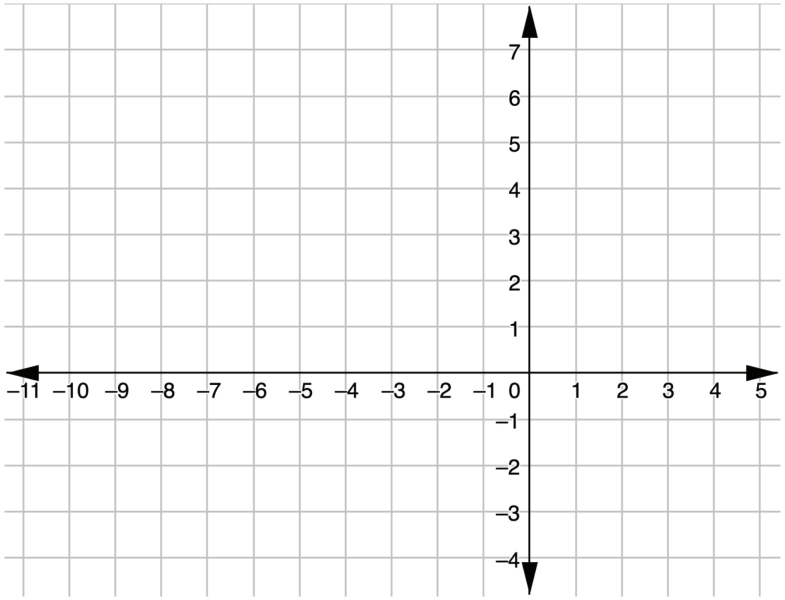
a. Show: ΔSCW is **NOT** an equilateral triangle



b. What type of triangle is ΔSCW?

3. Diego is creating a triangular garden and uses the ΔYVX with vertices at $Y(-8,-2)$, $V(-6,3)$, and $X(-2,-2)$ to design the garden.

a. Graph ΔYVX.

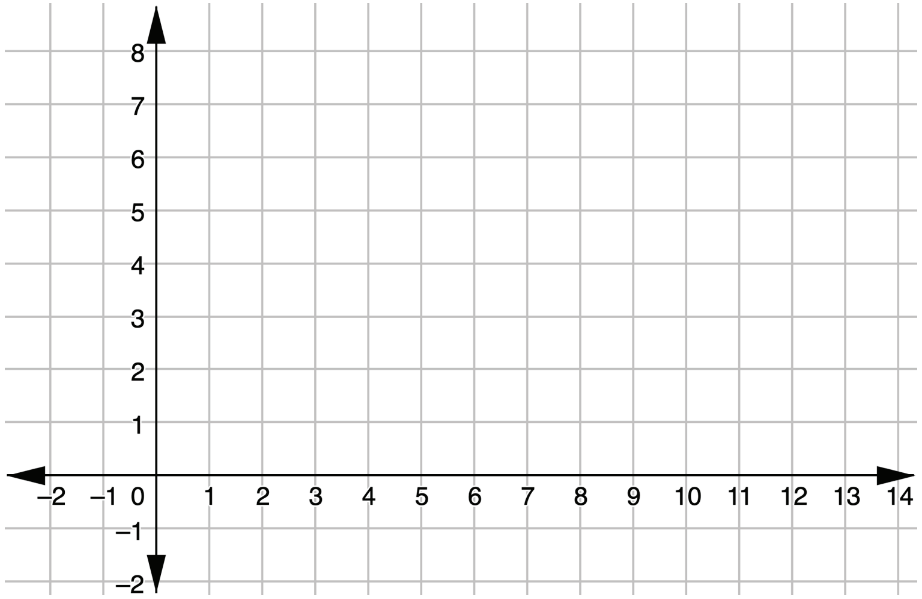


b. Complete the table. If necessary, round to the nearest thousandth.

Line Segment	**Side Length**
$\overline{YV}$	
$\overline{VX}$	
$\overline{XY}$	

c. Classify the type of triangular garden Diego is creating.

4. Draw an isosceles triangle whose base is three more than the height of the triangle and has one vertex at $(2, 5)$.



13.6.5 Wrap-Up: Cool Down

1. Identify **two** strategies you can use to classify triangles using coordinate geometry.

Unit 13, Lesson 7: Centroids of Triangles

Benchmark

MA.912.GR.3.3 Use coordinate geometry to solve mathematical and real-world geometric problems involving lines, circles, triangles, and quadrilaterals.

Learning Target

☐ I can use coordinate geometry to identify the centroid of a triangle.